

Replication Report: Phase–Amplitude Collective-Mode Mixing in the Imaginary CDW (Loop-Current) Phase of Kagome AV_3Sb_5

Automated replication (TEXTURES-100) of Xie & Nagaosa, arXiv:2504.14166

Verdict: **REPLICATED**

Abstract

We reproduce from scratch the central theoretical claim of Xie & Nagaosa (arXiv:2504.14166): in the imaginary charge-density-wave (iCDW, loop-current) phase of a triple- Q kagome CDW, the amplitude (Higgs) and phase collective modes *mix* in the A (C_3 -symmetric) irrep, whereas in the real-CDW (rCDW) reference they *decouple*. Starting only from the mean-field free energy (Eq. 2) and the fluctuation Lagrangian (Eq. 10), we minimize the free energy to obtain the equilibrium order parameter and the symmetry-selected phase θ_0 , build and diagonalize the $q = 0$ A -channel dynamical matrix, and recover the closed-form mixed-mode spectrum (Eq. 12) to machine precision ($\sim 4 \times 10^{-16}$). The off-diagonal phase–amplitude susceptibility is finite in the iCDW phase and vanishes (to $\sim 10^{-17}$, i.e. exactly by symmetry) in the rCDW phase, confirming the headline. A microscopic loop-current cross-check using a shared kagome tight-binding kernel confirms the underlying imaginary-hopping bond-current picture.

1 Model and method

The order parameter is the triple- Q CDW matrix $\Delta_{Q_j}(\mathbf{r}) = |\Delta_{Q_j}|e^{i(\mathbf{Q}_j \cdot \mathbf{r} + \theta_j)}$ on the kagome lattice (sublattices A, B, C ; ordering vectors $Q_{1,2,3}$ at the M points). With $\lambda_3 = 0$ and imposed C_3 symmetry ($\theta_1 = \theta_2 = \theta_3 \equiv \theta_0$, $|\Delta_{Q_1}| = |\Delta_{Q_2}| = |\Delta_{Q_3}| \equiv |\Delta_Q|$), the mean-field free energy (paper Eq. 2) reduces to

$$F = 3(b \cos 2\theta_0 - \lambda'_1)|\Delta_Q|^2 + \lambda_2|\Delta_Q|^3 \cos 3\theta_0 + (3u_1 + 6u_2)|\Delta_Q|^4. \quad (1)$$

The dominant quadratic term selects $\theta_0 = \pm\pi/2$ for $b > 0$ (iCDW) and $\theta_0 = 0, \pi$ for $b < 0$ (rCDW). We minimize F numerically over $(|\Delta_Q|, \theta_0)$.

The $q = 0$ A -channel fluctuation quadratic form, read directly from Eq. (10) in basis $(A^{(A)}, \theta^{(A)})$, is

$$M = \begin{pmatrix} 4(u_1 + 2u_2)|\Delta_Q|^2 & \frac{3}{2}\lambda_2 \sin(3\theta_0)|\Delta_Q| \\ \frac{3}{2}\lambda_2 \sin(3\theta_0)|\Delta_Q| & 2|b| - \frac{3}{2}\lambda_2 \cos(3\theta_0)|\Delta_Q| \end{pmatrix}, \quad (2)$$

and the mode energies satisfy $\det[M - \kappa_0\omega^2\mathcal{K}] = 0$. The off-diagonal element $\propto \sin(3\theta_0)$ is the mixing coefficient: nonzero for iCDW ($\sin(3\theta_0) = \mp 1$), exactly zero for rCDW ($\sin(3\theta_0) = 0$). For iCDW this yields the closed form (paper Eq. 12)

$$\kappa_0(\omega_{\pm}^{(A)})^2 = |b| + 2(u_1 + 2u_2)|\Delta_Q|^2 \pm \sqrt{(|b| - 2(u_1 + 2u_2)|\Delta_Q|^2)^2 + \frac{9}{4}\lambda_2^2|\Delta_Q|^2}. \quad (3)$$

2 Results

Parameters (paper Fig. 3): $\lambda'_1 = 5$, $\lambda_2 = 0.1$, $u_1 = u_2 = 0.5$; $b = +1$ (iCDW), $b = -1$ (rCDW).

Quantity	iCDW ($b = +1$)	rCDW ($b = -1$)
θ_0/π	0.5	1.0
$ \Delta_Q $ equilibrium	1.41421	1.42257
$ \Delta_Q $ analytic $\sqrt{(b + \lambda'_1)/2(u_1 + 2u_2)}$	1.41421	–
off-diag mixing coeff	–0.21213	7.8×10^{-17}
cross susceptibility $\chi_{A\theta}$	8.86×10^{-3}	-2.9×10^{-18}
$\kappa_0 \omega_-^2$ (numeric)	1.99550	2.21339
$\kappa_0 \omega_+^2$ (numeric)	12.00450	12.14226
Eq. (12) vs numeric max err	4.4×10^{-16}	–
modes mix?	yes	no

Signature. `mixing_present_in_iCDW_only = true`; `Eq12_reproduced = true`. The numerically diagonalized A -channel spectrum equals the paper’s closed-form Eq. (12) to machine precision, and the phase–amplitude cross susceptibility is finite for iCDW and zero (by symmetry) for rCDW.

Microscopic cross-check. Using the shared kagome loop-current mean-field kernel (`ollie_loop_current_meanfield_kernel.py`) the net loop order at zero Peierls flux is 0 (rCDW analog), while the loop-current susceptibility to an imaginary bond flux is finite (–62.2), consistent with the iCDW’s imaginary inter-sublattice hopping generating bond currents.

3 Verdict

REPLICATED. The from-scratch mean-field + RPA-style fluctuation calculation reproduces the paper’s qualitative headline (mode mixing in iCDW, decoupling in rCDW) *and* the quantitative closed-form spectrum Eq. (12) to machine precision. Not attempted: the NV-center T_1 relaxometry / stray-field magnitude estimate (Eqs. 13–15), which requires unspecified material/geometry inputs.

Credits

Microscopic loop-current cross-check uses the shared `loop_current_meanfield_kernel.py` (`ollie_loop_current_meanfield_kernel.py`) from `shared-kernels-cache/`. All numerical values are computed by `report/evidence/xie2025_replication.py` results in `work/xie2025_result.json`.